

CS & IT ENGINEERING



Computer Network

Error Control

Lecture No. - 07



By - Abhishek Sir



Recap of Previous Lecture



Topic

2D Parity

Topic

Hamming Code



Topics to be Covered



Topic

Hamming Code

Topic

Hamming Distance

[V. Imp]

ABOUT ME



Hello, I'm **Abhishek**

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Topic : 2D Parity



Minimum number of bits corrupted in the Block

= Maximum [total number of row wise parity unbalanced,
total number of column wise parity unbalanced]

#Q. Data transmitted on a link uses the following 2D parity scheme for error detection:

Each sequence of 28 bits is arranged in a 4×7 matrix (rows r_0 through r_3 , and columns d_7 through d_1) and is padded with a column d_0 and row r_4 of parity bits computed using the Even parity scheme. Each bit of column d_0 (respectively, row r_4) gives the parity of the corresponding row (respectively, column). These 40 bits are transmitted over the data link.

The table shows data received by a receiver and has n corrupted bits. What is the minimum possible value of n ?

[GATE 2008]

(A) 1

(B) 2

☒ (C) 3

(D) 4

Ans: C

Even Parity

AT Receiver: -



	d_7	d_6	d_5	d_4	d_3	d_2	d_1	d_0	
r_0	0	1	0	1	0	0	1	1	✓
r_1	1	1	0	0	1	1	1	0	←
r_2	0	0	0	1	0	1	0	0	✓
r_3	0	1	1	0	1	0	1	0	✓
r_4	1	1	0	0	0	1	1	0	✓
	✓	✓	↑ ①	✓	✓	↑ ②	✓	↑ ③	

one row wise parity is unbalanced.

Three column wise parity are unbalanced

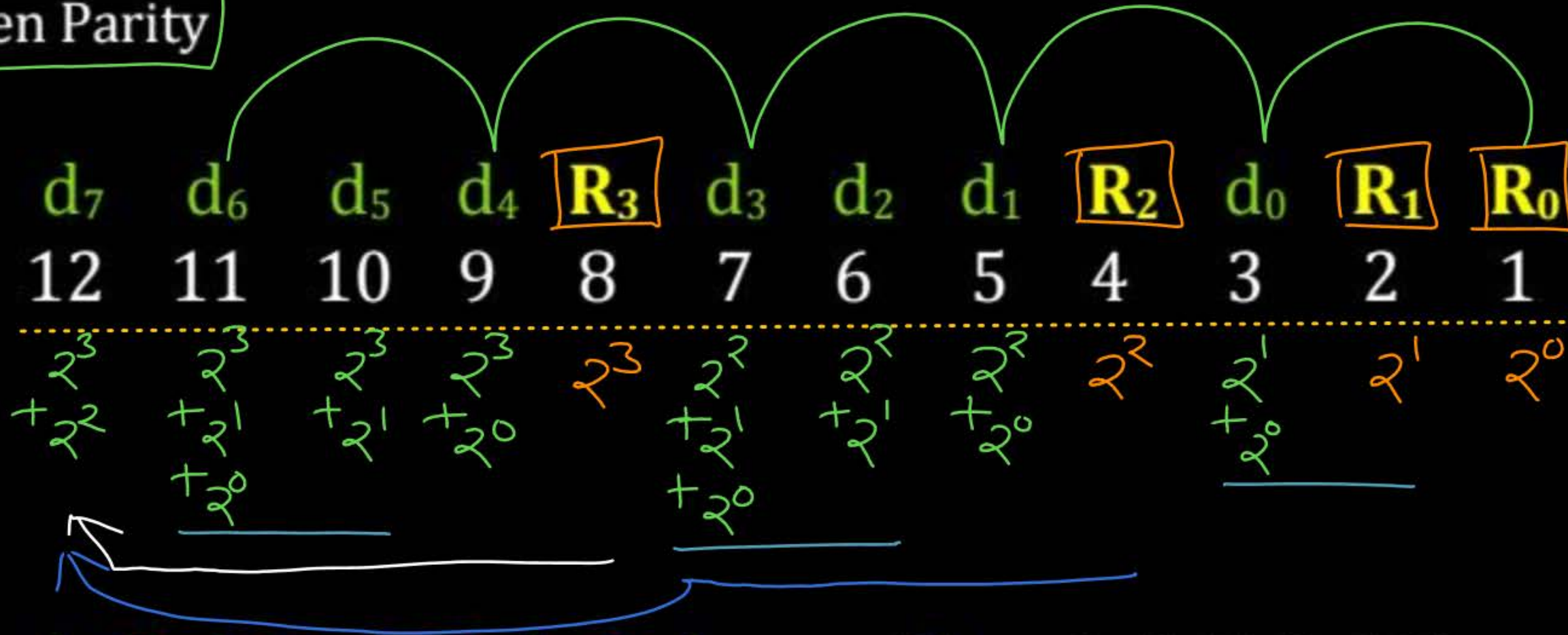
$$\max(1, 3) = 3$$



Topic : Hamming Code



→ Even Parity



$$R_0 = d_0 \oplus d_1 \oplus d_3 \oplus d_4 \oplus d_6$$

$$R_1 = d_0 \oplus d_2 \oplus d_3 \oplus d_5 \oplus d_6$$

$$R_2 = d_1 \oplus d_2 \oplus d_3 \oplus d_7$$

$$R_3 = d_4 \oplus d_5 \oplus d_6 \oplus d_7$$

$$[1 = 3, 5, 7, 9, 11]$$

$$[2 = 3, 6, 7, 10, 11]$$

$$[4 = 5, 6, 7, 12]$$

$$[8 = 9, 10, 11, 12]$$



Topic : Hamming Code



Example 2 :

Consider data bits are "0 0 1 1" and "Even Parity", generate hamming code?





Topic : Hamming Code



Example 2 :

Consider data bits are “0 0 1 1” and “Even Parity”, generate hamming code?

<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>0</u>
7	6	5	4	3	2	1

Ans : 0011110

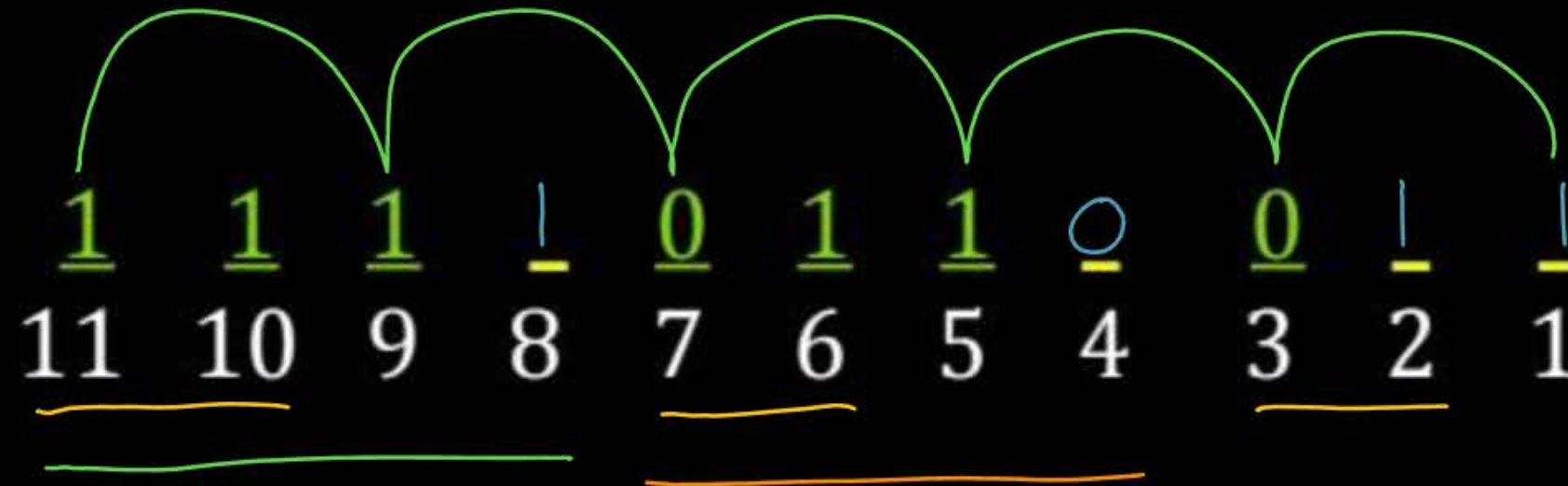


Topic : Hamming Code



Example 3 :

Consider data bits are "1110110" and "Even Parity", generate hamming code?





Topic : Hamming Code



Example 3 :

Consider data bits are “1110110” and “Even Parity”, generate hamming code?

<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>
11	10	9	8	7	6	5	4	3	2	1

Ans : 11110110011



Topic : Hamming Code



Example 4 :

Consider data bits are "10101101" and "Even Parity", generate hamming code?



R_3	R_2	R_1	R_0
0	1	0	0



Topic : Hamming Code



Example 4 :

Consider data bits are “10101101” and “Even Parity”, hamming code :

<u>1</u>	<u>0</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>
12	11	10	9	8	7	6	5	4	3	2	1

Ans : 101001101100



Topic : Hamming Code



At receiver :

CASE I : No any error

Transmitted Codeword = 1 0 1 0 0 1 1 0 1 1 0 0

Received Codeword = 1 0 1 0 0 1 1 0 1 1 0 0

<u>1</u>	<u>0</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>
12	11	10	9	8	7	6	5	4	3	2	1

Even Parity

P_0	✓
P_1	✓
P_2	✓
P_3	✓



Topic : Hamming Code



if receiver finds all the parity bits are balanced
then receiver concluded “No any error detected”

else
receiver concluded “Error detected”



Topic : Hamming Code



At receiver :

CASE I : No any error

<u>1</u>	<u>0</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>
12	11	10	9	8	7	6	5	4	3	2	1

Receiver Concluded : "No any error detected"

[Receiver finds all parity bits are balanced]

Error Position = 0

$R_3 R_2 R_1 R_0 = 0000$



Topic : Hamming Code



At receiver :

Even Parity

CASE II : One-bit error

Transmitted Codeword = 1 0 1 0 0 1 1 0 1 1 0 0

Received Codeword = 1 0 1 0 0 0 1 0 1 1 0 0

1	0	1	0	0	0	1	0	1	1	0	0
12	11	10	9	8	7	6	5	4	3	2	1

P_0 X
 P_1 X
 P_2 X
 P_3 ✓

$$7 = 4 + 2 + 1$$



Topic : Hamming Code



At receiver :

CASE II : One-bit error

Receiver Concluded : “Error detected”

[Receiver finds parity bits R_0, R_1 , and R_2 are unbalanced]

Error Position = 7 [$R_3 R_2 R_1 R_0 = 0 1 1 1$]

Corrected Codeword = 1 0 1 0 0 1 1 0 1 1 0 0

[Receiver successfully corrected single-bit error]



Topic : Hamming Code



Even Parity

At receiver :

CASE III : Two-bit error

Transmitted Codeword = 1 0 1 0 0 1 1 0 1 1 0 0

Received Codeword = 1 0 1 0 0 0 1 1 1 1 0 0

1	0	1	0	0	0	1	1	1	1	0	0
12	11	10	9	8	7	6	5	4	3	2	1

P_0 ✓
 P_1 X
 P_2 ✓
 P_3 ✓



Topic : Hamming Code



At receiver :

CASE III : Two-bit error

Receiver Concluded : “Error detected”

[Receiver finds only parity bits R_1 is unbalanced]

Error Position = 2 [R_3 R_2 R_1 R_0 = 0 0 1 0]

Corrected Codeword = 1 0 1 0 0 0 1 1 1 1 1 0

[Receiver can not correct burst error]



Topic : Hamming Code



At receiver :

CASE IV : Three-bit error

Transmitted Codeword = 1 0 1 0 0 1 1 0 1 1 0 0

Received Codeword = 1 0 1 0 0 1 1 0 1 0 1 1

1	0	1	0	0	1	1	0	1	0	1	1
12	11	10	9	8	7	6	5	4	3	2	1

R_0 ✓
 R_1 ✓
 R_2 ✓
 R_3 ✓



Topic : Hamming Code



At receiver :

CASE IV : Three-bit error

Receiver Concluded : “No any error detected”
[Receiver finds all parity bits are balanced]

[MCQ]IIT-B, H.W. **[GATE-2021][2 Mark]**

#Q. Assume that a 12-bit Hamming codeword consisting of 8-bit data and 4 check bits is $d_8d_7d_6d_5c_8d_4d_3d_2c_4d_1c_2c_1$, where the data bits and the check bits are given in the following tables:

Data bits							
d_8	d_7	d_6	d_5	d_4	d_3	d_2	d_1
1	1	0	x	0	1	0	1

Check bits			
c_8	c_4	c_2	c_1
y	0	1	0

Which of the following choices gives the correct values of x and y?

A

x is 0 and y is 0

B

x is 0 and y is 1

C

x is 1 and y is 0

D

x is 1 and y is 1



Topic : Hamming Code



→ Minimum number of parity bits required

$$2^r > (m + r)$$

→ Minimum code length = 3
[contains only one data bit and two parity bit]

→ Code length $\neq 2^i$
[where $i = 0, 1, 2, 3 \dots$]



Topic : Hamming Code



→ Using r parity bit,

$$\text{Maximum code length} = (2^r - 1)$$

$$\text{Maximum data bits} = (2^r - r - 1)$$

$$\rightarrow \text{Code rate} = \frac{(2^r - r - 1)}{(2^r - 1)}$$

$$\text{code rate} = \frac{\text{No. of data bits}}{\text{No. of data + Parity bits}}$$



Topic : Linear Code



Hamming Code (with Even Parity) and 3 data bits :

data = 011

0 1 1 1 1 0
6 5 4 3 2 1

Data ($d_2 d_1 d_0$)	-->	Codeword ($d_2 d_1 R_2 d_0 R_1 R_0$)
000	-->	000000
001	-->	000111
010	-->	011001
011	-->	011110
100	-->	101010
101	-->	101101
110	-->	110011
111	-->	110100

code belongs to Hamming
code Error correction method
are always linear code,
but not cyclic code.

2³

101010
101101

000111

→ } valid
→ } code
→ }


101010 ⇒ valid code
↓ one bit cyclic left shift

010101 → invalid code



Topic : Hamming Weight



→ The weight of a codeword is the number of nonzero elements

Hamming weight of a binary string = Number of one's in that string

Example :-

Hamming Weight["1011101"] = 5



Topic : Hamming Distance



→ Metric for comparing two binary strings of equal length

Hamming distance between two binary strings of equal length

= Number of positions at which the corresponding bits are different



Topic : Hamming Distance



→ Let suppose A and B are two binary strings of equal length

$$\begin{aligned} d(A, B) &= \text{Hamming Distance between } A \text{ and } B \\ &= \text{Hamming Weight of } [A \text{ bit-wise XOR } B] \end{aligned}$$

$$d(A, A) = 0$$



Topic : Hamming Distance



Example :- A = 1011101 and B = 0110110

bit-wise XOR A = 1 0 1 1 1 0 1
 B = 0 1 1 0 1 1 0

Result = 1 1 0 1 0 1 1

$d(A,B) = \text{Hamming Weight}[\text{Result}]$

$= \text{Hamming weight}["1101011"]$

$= 5$

$$A = 10101100$$

$$B = 01101101$$

(bitwise XOR)

$$\begin{array}{r} 10101100 \\ 01101101 \\ \hline 11000001 \end{array}$$

$$d(A, B) = ? = 3$$



Topic : Hamming Distance



Let suppose set of valid codewords :

Codeword C_1 : 00000 00000

Codeword C_2 : 00000 11111

Codeword C_3 : 11111 00000

Codeword C_4 : 11111 11111

minimum Hamming Distance =

Minimum[$d(C_1, C_2)$, $d(C_1, C_3)$, $d(C_1, C_4)$, $d(C_2, C_3)$, $d(C_2, C_4)$, $d(C_3, C_4)$]

H.W.



2 mins Summary



Topic

Hamming Code ✓

Topic

Hamming Distance

[v. Imp.]



THANK - YOU

